

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 9 Lecture - Integrating Second-Order ODEs

Last Week...

- In some cases, we cant directly integrate by using the common laws, as we are faced with more complicated combinations of functions such as **functions of functions**.
- For example, what happens if we want to integrate

$$y = \int \frac{dy}{dx} = \int (3x + 2)^4 dx$$

- This is where **integration by substitution** comes in.
- Integration by substitution is essentially the reverse process of the chain rule in differentiation.

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

Last Week...

1. Write out the integral in its simplest form:

$$y = \int (3x + 2)^4 dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x + 2$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u - 2)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. (If needed) Find the new limits of the integral:

$$\text{When } x = -1, \quad u = 3 \times (-1) + 2 = -1$$
$$\text{When } x = 1, \quad u = 3 \times 1 + 2 = 5$$

7. Make the substitution in the original integral

$$\frac{1}{3}\int_{-1}^{5} u^4 \, du = \frac{1}{3}\left[\frac{1}{5}u^5\right]_{-1}^{5} = \frac{1}{3} \times \frac{1}{5} \times (5^5 - 1^5) = \frac{3124}{15}$$

8. Integrate as usual.

Last Week...

- Hydrostatic Equation, with constant temperature requires integration via separation of variables:

$$\frac{dP}{dz} = -\frac{g}{R_\alpha T}P \quad \Longrightarrow \quad \int \frac{1}{P}dP = \int -\frac{g}{R_\alpha T}dz$$
$$\Longrightarrow \quad P(z) = P_0 \exp\left(-\frac{g}{R_\alpha T}z\right)$$

- Hydrostatic Equation, with varying temperature requires integration by substitution:

$$\frac{dP}{dz} = -\frac{P}{R_a\left(T_{surface} - 0.01z\right)}g \quad \Longrightarrow \quad \int \frac{1}{P}dP = -\frac{g}{R_a}\int \frac{1}{T_{surface} - 0.01z}dz$$
$$\Longrightarrow \quad P(z) = P_0 \exp\left(\frac{100g}{R_a}\ln\left(\frac{T_{surface} - 0.01z}{T_{surface}}\right)\right)$$

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Describe the concept of second-order ordinary differential equations and explain their role in mathematical and physical modelling.

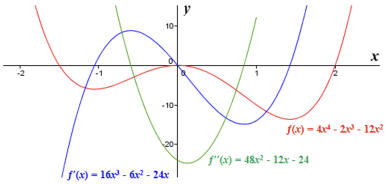
(2) Solve second-order ODEs to generate general solutions using the auxiliary equation method

(3) Determine full solutions to second-order ODEs by applying given initial conditions.

(4) Apply second-order ODEs to model physical processes through boundary conditions in real-world examples such as Newtons Laws, Heat Transfer, and Mass Transfer

Second-order Differential Equations

- We started thinking about ODEs from the need to calculate a gradient.
- In Week 5, we discovered that differentiating a function twice can also be useful.
- For a stationary value x for the function $f(x)$:
  - If $f''(x) < 0$ the stationary value is a maximum
  - If $f''(x) > 0$ the stationary value is a minimum
  - If $f''(x) = 0$ it is not possible to conclude anything about the stationary value.
- If we're in the situation that $f''(x) = 0$, then it is potentially a saddle point. Need to investigate further.



Second-order Differential Equations

- We can write a physical law using second order ODEs and we have implicitly done this already for Newtons second law.
- In Week 5, we discussed second-order derivatives and observed that acceleration (a) is the rate of change of velocity (v), while velocity (v) is the rate of change of position or distance (x).
- In Week 7, we showed that all objects fall the same distance (quadratically) over the same amount of time in a vacuum, regardless of their mass.

$$\begin{aligned} \frac{dv}{dt} &= g \implies v(t) = gt \\ \frac{dx}{dt} &= gt \implies x(t) = \frac{1}{2}gt^2 \end{aligned}$$



- So... we have written and solved a second-order ODE!

Second-order Differential Equations

- **Second-order ordinary differential equations** are ODEs that involve a **second-order derivative**.
- We could devote an entire course looking at solving different types of second-order ODEs, but we will only concentrate on

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

where a, b and c are constants.

- This is referred to as a **homogeneous, linear, second-order ODE**.
- Note that we can write the exact same expression using alternative notation as per the following equation:

$$ay'' + by' + cy = 0$$

Solving Second-order Differential Equations

- In every problem so far, we have wanted to arrive at a general analytical solution from integrating a differential equation.
- Fortunately, there is a general form we can use if the second-order ODE to be solved has the same form.
- It looks like we might be able to use the approach to solving quadratic equations, but how do we find a value for y?
- Let's just jump to an educated guess and assume the expression for y is going to be:

$$y = e^{\lambda x}$$

where λ is a constant with an unknown value.

Solving Second-order Differential Equations

- If we differentiate (twice) this expression we can arrive at two expressions:

$$\frac{dy}{dx} = \lambda e^{\lambda x} \qquad \frac{d^2y}{dx^2} = \lambda^2 e^{\lambda x}$$

- Lets now substitute these expressions back into our general differential equation as follows:

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Because $e^{\lambda x}$ cant be zero, it is safe to divide through to acquire a quadratic in λ:

$$a\lambda^2 + b\lambda + c = 0$$

- This is called the **auxiliary** equation or **characteristic** equation.
- When you encounter a **homogeneous second-order ODE**, you can jump straight to this point.

Solving Second-order Differential Equations

- Looks like we have a quadratic equation.
- If we can solve this, we get a value of λ which we can use to find a solution to the second-order ODE.
- Remember that we can solve (or find roots of) quadratic equations using the following formula:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- When we solve a quadratic expression we will arrive at 2 roots, which we can call λ_1 and λ_2.
- Since there are two roots to the auxiliary equation, the general solution to the second-order ODE will be a combination of the two functions.

Solving Second-order Differential Equations

- If λ_1 and λ_2 are real and not equal ($\lambda_1 \neq \lambda_2$ then we can write a general solution as:

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

- If our two roots are real and equal (i.e. $\lambda_1 = \lambda_2$) then we use the formula:

$$y = (A + Bx)e^{\lambda_1 x}$$

- We don't consider what happens when the two roots are not real (i.e. $b^2 - 4ac < 0$) in this course.

Example: Second-order Differential Equations

- Imagine we are asked to solve the following linear, homogeneous, second-order ODE:

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 5y = 0$$

- Firstly, we write out the auxiliary equation:

$$\lambda^2 + 6\lambda + 5 = 0$$

- Secondly, find the roots (λ_1 and λ_2) of this equation to understand what solution is needed:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{6 \pm \sqrt{6^2 - 4 * 1 * 5}}{2 * 1} = \frac{6 \pm 4}{2} = 3 \pm 2$$

- So the roots are $\lambda_1 = 1$ and $\lambda_2 = 5$.

Example: Second-order Differential Equations

- Since they are real and distinct then the solution will have form

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

- Finally, we plug the values of λ_1 and λ_2

$$y = Ae^{x} + Be^{5x}$$

and this is our general solution to the problem.

Solving Second-order Differential Equations

- The next question is how to calculate the values for constants A and B?
- We need more information to constrain these values such as $y = 0$ and $y = 1$ when $x = 0$.
- These are called **initial conditions**, and they tell us the value of the solution and/or its derivatives at a specific point, allowing for a unique solution to the differential equation.
- For example, for $\frac{dy}{dx} = f(x)$, an initial condition might be $y = y_0$ at $x = x_0$.
- You will have seen examples when solving the radioactive decay equation, hydrostatic equation and Newton's Laws of Motion.
- This will lead us to solving simultaneous equations.

Example: Second-order Differential Equations

- Let's return to the previous example:

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 5y = 0$$

- The general solution to the problem was

$$y = Ae^x + Be^{5x}$$

- Suppose that we are told that $y = 0$ and $y' = 1$ when $x = 0$ then we use this information to find A and B.
- Write out y and y':

$$\begin{aligned} y &= Ae^x + Be^{5x} \\ y' &= Ae^x + 5Be^{5x} \end{aligned}$$

Example: Second-order Differential Equations

- Plug in the initial conditions in to y and y' to get:

$$\begin{aligned} y(0) &= A + B = 0 \\ y'(0) &= A + 5B = 1 \end{aligned}$$

- This leaves us with a system of simultaneous equations which we can solve:

$$A + 5B - (A + B) = 1 - 0 \implies 4B = 1 \implies B = \frac{1}{4}$$

- Substitute $B = \frac{1}{4}$ into $A + B = 0$ gives us:

$$A + \frac{1}{4} = 0 \implies A = -\frac{1}{4}$$

- Then the full solution is given by:

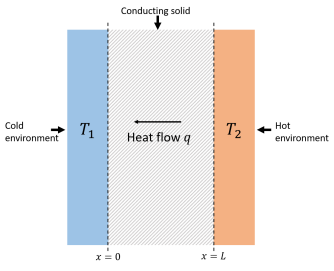
$$y = \frac{1}{4}e^{5x} - \frac{1}{4}e^{x}$$

Heat Transfer and Fourier's Laws

- Heat moves from hot to cold regions and the material, or medium, properties dictate the rate of heat transfer.
- **Fourier's Second Law**, describes how the distribution of heat (temperature) in a material changes over time:

$$\frac{dT}{dt} = \alpha \frac{d^2T}{dx^2}$$

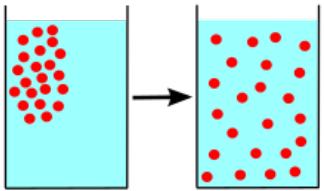
  where α is the thermal diffusivity (speed heat diffuses).
- Thermal diffusivity combines the material's thermal conductivity, density, and specific heat capacity.



- Fick also found that the law that governs transport of mass by Brownian motion (i.e. random thermal motion) is very similar (i.e. mass moves from high to low concentrations).
- **Fick's Second Law**, takes into account the time-dependent change in concentration within a material and the principle of conservation of mass:

$$\frac{dC}{dt} = D\frac{d^2C}{dx^2}$$

where D is a diffusion coefficient which depends on the properties of the substance that is diffusing



Fourier's and Fick's laws

- Remember that a system is in steady-state when the rate of change that describes the system's behavior do not change over time.
- If we assume steady state, we get second-order ODEs which we can simply integrate twice

$$\frac{d^2T}{dx^2} = 0 \implies T(x) = Ax + B$$
$$\frac{d^2C}{dx^2} = 0 \implies C(x) = Cx + D$$

- We can use boundary conditions to solve this and find an expression.
- **Boundary value problems** differ from **initial value problems** as we specify upper and lower boundary conditions are specified (i.e, when $x = x_0$ and $x = x_1$).
- This gives us more information this means that we have to make the solution meet at both points.